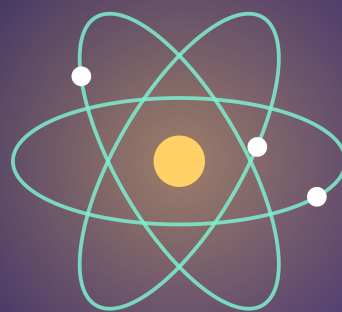
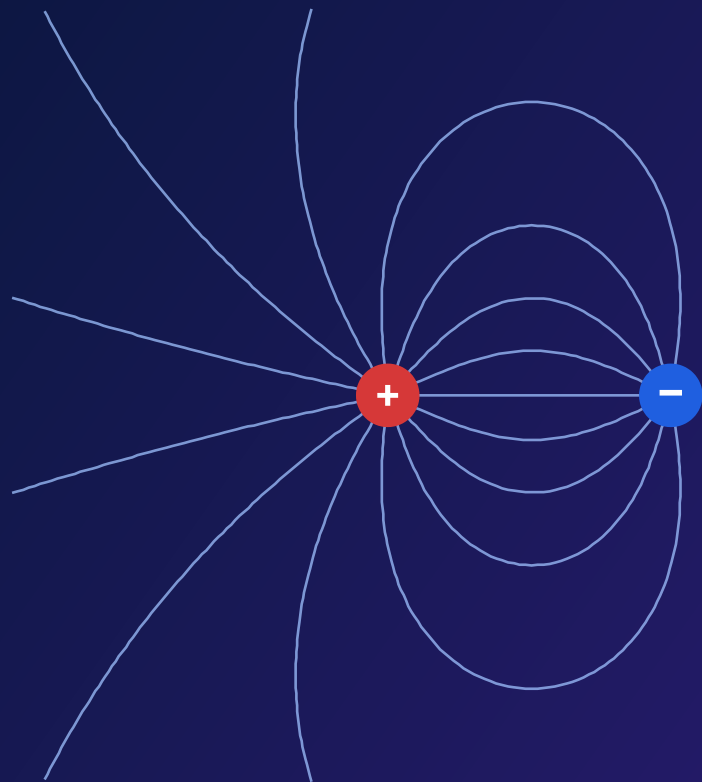




CLASS 12 · 2026-27

CBSE PHYSICS SCORE GUIDE

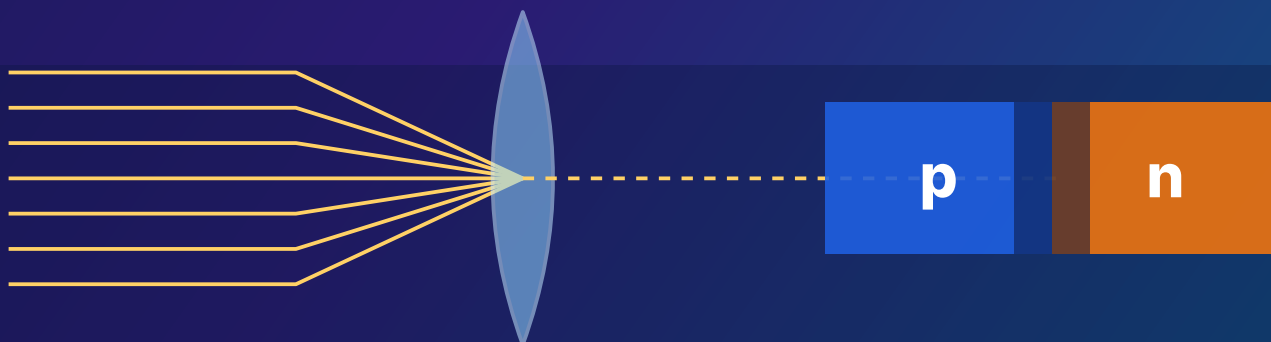
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About the author



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VS Sir has taught Physics to NEET and IIT-JEE aspirants of Class 11, Class 12 and droppers in Kota for more than 20 years, in English and Hindi medium. He is the founder of VCA Kota and the author of the *NEET + JEE Physics 30-Day Rapid Revision* books, *Electrostatics* and *Practical Guide to AI for Educators*. His method is simple: understand the idea, draw the diagram, then practise until the steps become automatic.

Doubts and feedback: WhatsApp +91 96024 05311

A message to the student

DEAR STUDENT,

Physics is not a subject to be memorised; it is a subject to be **seen**. Every chapter in this book is built on one habit: *draw it, name it, write the formula, then solve*. Most marks in the board paper are won by clear diagrams, correct formulas with units, and answers written in the steps an examiner expects.

Use this book with a pencil. Cover the green answer strips, try each question first, then compare. Mark what you got wrong in the revision page of each chapter and return to it before the exam. A little, every day, beats a lot, once.

— VS Sir

How to use this book

1. LEARN

Read the chapter opener (outcomes, prerequisites, exam note), then the concept pages. Look at every labelled diagram.

2. DERIVE & SOLVE

Practise each derivation (principle, steps, result). Solve numericals N1–N6 before reading the working.

3. TEST

Attempt MCQs and subjective questions with the answer strip covered. Note the question ID of anything you miss.

Colour code used throughout

● Blue	Core concept and definition
● Violet	Advanced / extension idea
● Teal	Correct method, checkpoint
● Orange	Formula, derivation, exam tip
● Red	Common mistake — avoid
● Gold	Revision and summary

Syllabus coverage checklist

Tick each chapter when you have finished concepts, derivations, numericals, MCQs and the revision page. Chapter names follow the usual Class 12 NCERT Physics sequence; **confirm the exact 2026–27 syllabus and any deleted or added topics on the official board website.**

CHAPTER	UNIT	MAIN TOPICS	PAGE
☐ ● 1. Electric Charges and Fields	Electrostatics	Coulomb's law, field, dipole, Gauss's law	8
☐ ● 2. Electrostatic Potential and Capacitance	Electrostatics	Potential, dipole, conductors, capacitors, dielectrics	17
☐ ● 3. Current Electricity	Current Electricity	Drift, Ohm, cells, Kirchhoff, Wheatstone bridge, meter bridge, potentiometer	26
☐ ● 4. Moving Charges and Magnetism	Magnetic Effects of Current and Magnetism	Biot–Savart, Ampere, force, cyclotron, torque, galvanometer	35
☐ ● 5. Magnetism and Matter	Magnetic Effects of Current and Magnetism	Bar magnet, Earth's magnetism, magnetic materials	44
☐ ● 6. Electromagnetic Induction	Electromagnetic Induction and AC	Faraday, Lenz, motional emf, inductance, eddy currents	53
☐ ● 7. Alternating Current	Electromagnetic Induction and AC	AC circuits, LCR, resonance, power, transformer	62
☐ ● 8. Electromagnetic Waves	Electromagnetic Waves	Displacement current, EM spectrum	71
☐ ● 9. Ray Optics and Optical Instruments	Optics	Reflection, refraction, lens, prism, instruments	78
☐ ● 10. Wave Optics	Optics	Huygens, interference, diffraction (polarisation: check)	89
☐ ● 11. Dual Nature of Radiation and Matter	Dual Nature	Photoelectric effect, Einstein's equation, de Broglie	98
☐ ● 12. Atoms	Atoms and Nuclei	Rutherford, Bohr model, hydrogen spectrum	106
☐ ● 13. Nuclei	Atoms and Nuclei	Size, binding energy, fission and fusion (decay: check)	114
☐ ● 14. Semiconductor Electronics	Electronic Devices	Bands, p–n junction, diode, rectifier	123

★ TOPICS MARKED “CHECK”

Polarisation (Chapter 10), radioactive decay (Chapter 13) and special-purpose diodes (Chapter 14) are included as clearly marked **extension** boxes because their examinable status for 2026–27 could not be verified. If the official syllabus excludes them, skip those boxes.

Syllabus at a glance

1 Electric Charges and Fields	2 Electrostatic Potential and Capacitance	3 Current Electricity	4 Moving Charges and Magnetism	5 Magnetism and Matter	6 Electromagnetic Induction	7 Alternating Current
8 Electromagnetic Waves	9 Ray Optics and Optical Instruments	10 Wave Optics	11 Dual Nature of Radiation and Matter	12 Atoms	13 Nuclei	14 Semiconductor Electronics

Exam roadmap and revision strategy

★ PAPER PATTERN SNAPSHOT — PROVISIONAL

Recent CBSE Class 12 Physics theory papers are **70 marks, 3 hours**, with objective items (including assertion–reason and case-based questions), short-answer and long-answer questions. The exact section-wise count and internal choices for 2026–27 **must be confirmed from the official sample paper**; this book could not verify them and does not rely on them. The 3 practice papers at the end follow the recent pattern and are original, not past papers.

A 12-week roadmap

Weeks 1–3

Ch 1–3. Concepts + derivations + numericals

Weeks 4–6

Ch 4–7. Magnetism, EMI, AC

Weeks 7–9

Ch 8–11. EM waves, optics, dual nature

Weeks 10–11

Ch 12–14 + mixed MCQs

Week 12

Three full papers, formula sheets, diagrams

F FORMULA REVISION

- Maintain one A4 sheet per chapter (use the revision pages).
- Say each formula aloud with units and conditions.
- Test: write 10 formulas from memory each day; check, correct, repeat.
- Never memorise a formula without the situation it applies to.

I DIAGRAM REVISION

- Use the “diagrams to practise” list on each revision page.
- Draw each in 2 minutes with labels and arrows.
- Mark ray directions, field directions and units.
- Redraw missed ones next day.

★ LAST 7 DAYS

- Day 1–3: formula sheets and revision maps (back section).
- Day 4–6: one full paper per day under timed conditions.
- Day 7: light revision of mistakes list; sleep well.

! IN THE EXAM HALL

- Read all questions; start with those you are surest of.
- Show formula, substitution, answer with unit.
- Label diagrams and draw with a pencil.
- Do not leave case-based questions blank — read the passage twice.

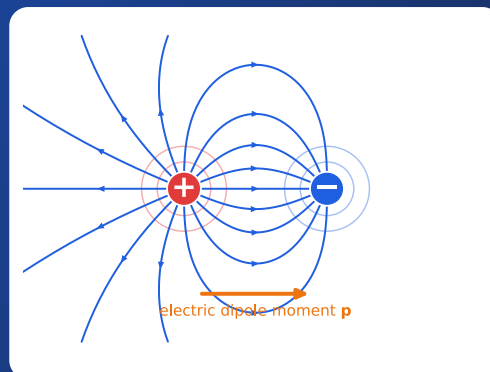
~ TIME PLAN FOR A 3-HOUR PAPER (GUIDE ONLY)

Reading: 10 min · objective: 25 min · short answers: 55 min · long answers and case-based: 70 min · revision: 20 min. Adjust once you know the official section pattern.

UNIT 1 · ELECTROSTATICS

1 Electric Charges and Fields

Everything in electrostatics starts here: how charges push and pull, how a field carries that push through space, and how symmetry lets Gauss's theorem find fields in one line.



Chapter roadmap

Charge & Coulomb's law

Electric field & field lines

Electric dipole

Flux & Gauss's theorem

Applications of Gauss

✓ LEARNING OUTCOMES

- Explain quantisation and conservation of charge and apply Coulomb's law in vector form
- Use the superposition principle for discrete and continuous charge distributions
- Sketch and interpret electric field lines; state their properties
- Derive the field of an electric dipole (axial and equatorial) and the torque on it
- State Gauss's theorem and apply it to a long wire, an infinite sheet and a thin spherical shell

I PREREQUISITES TO REVISE

- Vectors: addition, resolution, unit vector, dot and cross product
- Newton's laws; force as a vector
- Basic integration: $\int x^n dx$
- Area of a sphere $4\pi r^2$, area of a cylinder $2\pi r l$

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: electric charges; conservation of charge; Coulomb's law — force between two point charges, forces between multiple charges; superposition principle and continuous charge distribution; electric field, field due to a point charge, field lines; electric dipole, field due to a dipole, torque on a dipole in a uniform field; electric flux; **statement** of Gauss's theorem and its applications to an infinitely long straight wire, a uniformly charged infinite plane sheet and a uniformly charged thin spherical shell (inside and outside).

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs on Coulomb's law and field lines; 2-mark short answers (properties of field lines, dipole moment); 3-mark derivations (dipole field, torque, a Gauss's-law application); 5-mark long answers (axial/equatorial field or two Gauss's-law results with figures). *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

How this chapter connects to the rest of the syllabus

Ch 1

Field of fixed charges

Ch 2

Potential, capacitors: energy of the same charges

Ch 3

Charges in motion: current

Ch 4

Moving charges: magnetic force

F CONSTANTS USED IN THIS CHAPTER

Constant	Symbol	Value
Elementary charge	e	$1.602 \times 10^{-19} \text{ C}$
Coulomb constant	$k = \frac{1}{4\pi\epsilon_0}$	$9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Permittivity of free space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67 \times 10^{-27} \text{ kg}$

I THREE IDEAS TO CARRY FORWARD

- **Vector thinking:** forces and fields add as vectors, so direction is half the answer.
- **Symmetry:** choose the Gaussian surface (or components) so that the integral becomes one multiplication.
- **Fields are real:** the field stores and transmits the interaction — the idea behind capacitors, waves and light.

1.1 Charge and Coulomb's law

I FOUR PROPERTIES OF ELECTRIC CHARGE

- **Two kinds:** positive and negative. Like charges repel, unlike charges attract.
- **Additive:** total charge is the *algebraic* sum (charge is a scalar).
- **Conserved:** in an isolated system the net charge never changes; charge is only transferred.
- **Quantised:** $q = \pm ne$, $n = 1, 2, 3, \dots$, with $e = 1.602 \times 10^{-19}$ C. Because $1 \mu\text{C} \approx 6.24 \times 10^{12} e$, quantisation is invisible at the macroscopic scale.
- **Everyday link:** a rubbed comb attracts paper scraps because it polarises them — the nearer (opposite) charge is pulled harder than the farther (like) charge.

F COULOMB'S LAW

The force between two **point charges at rest** is directly proportional to the product of the charges and inversely proportional to the square of the distance; it acts along the line joining them.

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} = k \frac{|q_1 q_2|}{r^2}$$

$$k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}, \quad \epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}. \text{ In a medium: } F = \frac{F_{\text{vacuum}}}{\epsilon_r}.$$

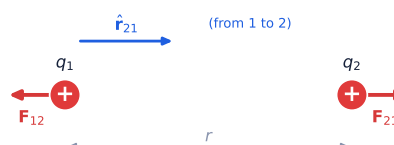


Fig. 1.1 Vector form: force on charge 2 due to charge 1 is $\mathbf{F}_{21} = k \frac{q_1 q_2}{r^2} \hat{\mathbf{r}}_{21}$. For like charges it points along $\hat{\mathbf{r}}_{21}$ (repulsion). $\mathbf{F}_{12} = -\mathbf{F}_{21}$ (Newton's third law).

1.2 Superposition and continuous charge

I PRINCIPLE OF SUPERPOSITION

The force on a charge due to several others is the **vector sum** of the forces due to each one separately; the other charges do not alter the pair-wise force.

$$\mathbf{F}_1 = \mathbf{F}_{12} + \mathbf{F}_{13} + \dots + \mathbf{F}_{1n} = \frac{q_1}{4\pi\epsilon_0} \sum_{j=2}^n \frac{q_j}{r_{1j}^2} \hat{\mathbf{r}}_{j1}$$

Method: (i) draw each force with its true direction, (ii) resolve into x, y components, (iii) add components, (iv) find magnitude and angle.

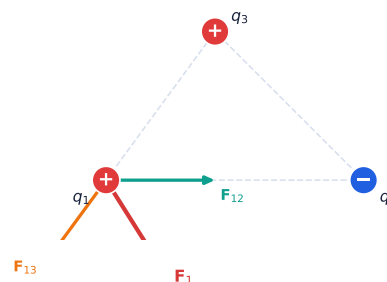


Fig. 1.2 Forces on q_1 : q_2 is negative (attracts q_1), q_3 is positive (repels q_1). The resultant $\mathbf{F}_1$ is the vector sum, not the sum of magnitudes.

~ CONTINUOUS CHARGE DISTRIBUTIONS (PRESCRIBED)

Treat the body as many tiny point charges dq ; add (integrate) their contributions vectorially. Use symmetry first — components that cancel by symmetry need not be integrated.

Distribution	Density	Charge element	Field element
Line (wire, ring)	$\lambda = \frac{dq}{dl} \text{ (C m}^{-1}\text{)}$	$dq = \lambda dl$	$d\mathbf{E} = \frac{dq}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}$
Surface (sheet, disc)	$\sigma = \frac{dq}{dA} \text{ (C m}^{-2}\text{)}$	$dq = \sigma dA$	integrate over the surface
Volume (solid sphere)	$\rho = \frac{dq}{dV} \text{ (C m}^{-3}\text{)}$	$dq = \rho dV$	integrate over the volume

★ HOW STRONG IS ELECTRICITY?

For an electron and proton:

$$\frac{F_{\text{electric}}}{F_{\text{gravity}}} = \frac{ke^2}{Gm_e m_p} \approx 2.3 \times 10^{39}. \text{ That is why gravity is negligible between charged particles but dominates for planets (which are neutral).}$$

! COMMON MISTAKES

- Using r in cm in $F = kq_1 q_2 / r^2$ — always convert to metres.
- Adding force *magnitudes* instead of vectors in superposition.
- Substituting signs of charges into the *magnitude* formula and then getting a negative 'magnitude'. Use signs only to decide direction.
- Applying Coulomb's law to extended bodies without treating them as point charges.

1.3 Electric field and field lines

F ELECTRIC FIELD

Field at a point = force per unit *test charge* (taken vanishingly small so that it does not disturb the source):

$$\mathbf{E} = \lim_{q_0 \rightarrow 0} \frac{\mathbf{F}}{q_0} \quad \mathbf{E}_{\text{point}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

Unit: N C^{-1} ($= \text{V m}^{-1}$). Force on charge q in a field: $\mathbf{F} = q\mathbf{E}$ — along $\mathbf{E}$ for $q > 0$, opposite for $q < 0$. For many charges:

$\mathbf{E} = \sum \mathbf{E}_i$ (vector sum).

I PROPERTIES OF ELECTRIC FIELD LINES

1. Start on positive charges, end on negative charges (or at infinity).
2. The **tangent** at any point gives the direction of $\mathbf{E}$ there.
3. Closer lines mean stronger field: density $\propto |\mathbf{E}|$.
4. Two field lines never cross — $\mathbf{E}$ would then have two directions at one point.
5. They do not form closed loops in electrostatics.
6. They meet a conductor's surface at right angles.
7. They are a visual aid; the field exists everywhere, not only on the lines.

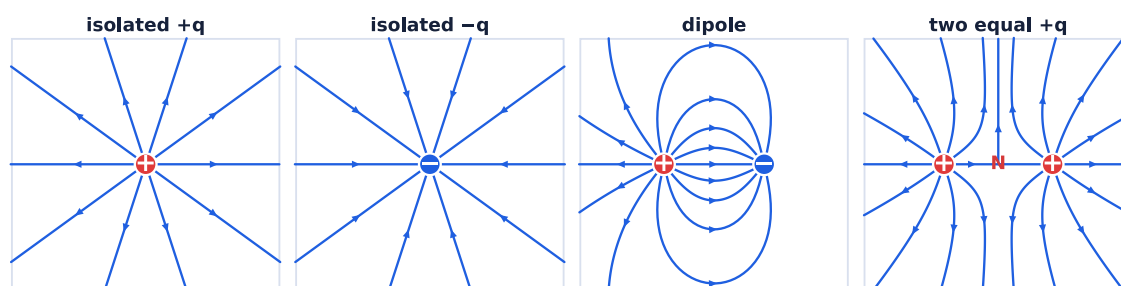


Fig. 1.3 Field-line patterns. For a lone $+q$ the lines point radially outward, for a lone $-q$ radially inward. The dipole pattern curves from $+q$ to $-q$. Between two equal $+q$ charges there is a **neutral point N** where the field is zero (lines repel and do not enter this region).

1.4 The electric dipole

I DEFINITION

Two equal and opposite charges $\pm q$ separated by a small distance $2a$. The **dipole moment** is a vector:

$$\mathbf{p} = q (2\mathbf{a}) \quad p = 2aq$$

Direction: from $-q$ to $+q$. SI unit: C m . Example: a water molecule has $p \approx 6.2 \times 10^{-30} \text{ C m}$. A *point (ideal) dipole* is the limit $a \rightarrow 0$, $q \rightarrow \infty$ with p fixed.

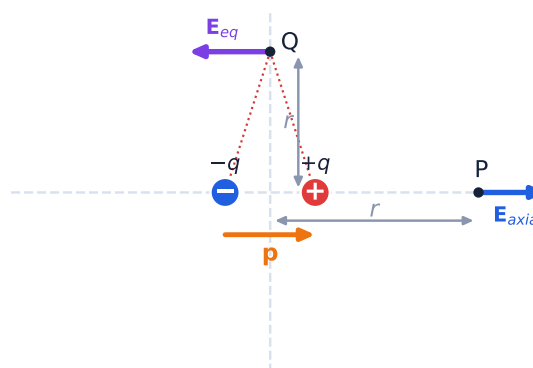


Fig. 1.4 Geometry used for the dipole field. **P** lies on the axial line, **Q** on the equatorial (perpendicular bisector) line, both at distance r from the centre. $\mathbf{E}_{\text{axial}}$ is along $\mathbf{p}$; $\mathbf{E}_{\text{eq}}$ is opposite to $\mathbf{p}$.

★ RESULTS TO REMEMBER (DERIVED ON THE NEXT PAGE)

Position	Exact	Short dipole ($r \gg a$)	Direction
Axial	$\frac{2pr}{4\pi\epsilon_0 (r^2 - a^2)^2}$	$\frac{2p}{4\pi\epsilon_0 r^3}$	along $\mathbf{p}$
Equatorial	$\frac{p}{4\pi\epsilon_0 (r^2 + a^2)^{3/2}}$	$\frac{p}{4\pi\epsilon_0 r^3}$	opposite to $\mathbf{p}$

Dipole field falls as $1/r^3$ (faster than a single charge's $1/r^2$) because the two fields partly cancel. Axial field is **twice** the equatorial field at the same r .

✓ CONCEPT CHECK (ANSWERS IN BRACKETS)

1. A positive test charge is placed at a point and feels a force $\mathbf{F}$. What is the force on an electron placed at the same point? [$-\mathbf{eE}$: opposite to $\mathbf{F}$ in direction]
2. Why can the electric field of a dipole at large distance not be $\propto 1/r^2$? [the two fields nearly cancel, leaving $1/r^3$]
3. A dipole is neutral. Does it feel a net force in a *uniform* field? [No — only a torque]

1.5 Derivations: dipole field and torque

Field of a dipole on the axial line

DERIVATION

Principle Superposition of the fields of $+q$ and $-q$.

Assumptions Short dipole; point P on the axis at distance r from the centre, $r > a$ (see Fig. 1.4).

- Distance of P from $+q$ is $(r - a)$ and from $-q$ is $(r + a)$.
- Field of $+q$ (away from $+q$, i.e. along **p**): $E_+ = \frac{1}{4\pi\epsilon_0} \frac{q}{(r - a)^2}$.
- Field of $-q$ (toward $-q$, i.e. opposite to **p**): $E_- = \frac{1}{4\pi\epsilon_0} \frac{q}{(r + a)^2}$.

Since $E_+ > E_-$, the resultant is along **p**.

- $E = E_+ - E_- = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(r - a)^2} - \frac{1}{(r + a)^2} \right] = \frac{q}{4\pi\epsilon_0} \frac{4ar}{(r^2 - a^2)^2}$
- Put $2aq = p$, so $4aqr = 2pr$: $E = \frac{1}{4\pi\epsilon_0} \frac{2pr}{(r^2 - a^2)^2}$. For $r \gg a$,
 $(r^2 - a^2)^2 \approx r^4$.

FINAL RESULT

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3} \quad (\text{along } \mathbf{p})$$

Limits: valid for a short dipole ($r \gg a$); exact form above is valid for any $r > a$ on the axis.

How to write it in the exam: Draw the dipole with P marked, show E_+ and E_- with their directions, write the four steps above and box the final result. (Typically: diagram 1 + working 1.5–2 + result 0.5–1 mark.)

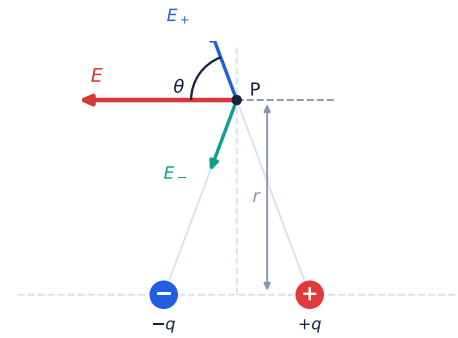


Fig. 1.5 Equatorial geometry: θ is the angle between E_+ and the axis, so the net field is $2E_+ \cos \theta$ opposite to **p**.

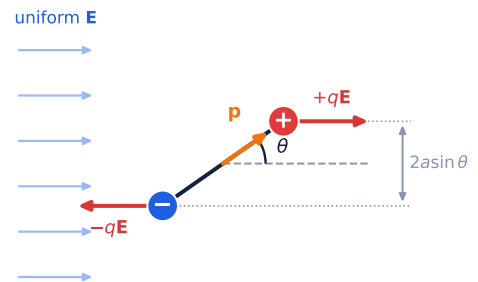


Fig. 1.6 Torque on a dipole in a uniform field. The two forces $\pm qE$ form a couple; their lines of action are $2a \sin \theta$ apart.

! SIGN CHECK

On the axial line the resultant is along **p**; on the equatorial line it is **opposite** to **p**. Both statements are asked often.

Field of a dipole on the equatorial line

DERIVATION

Principle Superposition; components perpendicular to the axis cancel.

Assumptions P on the perpendicular bisector at distance r from the centre; $d = \sqrt{r^2 + a^2}$ from each charge.

- Magnitudes are equal: $E_+ = E_- = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2 + a^2}$ (E_+ away from $+q$, E_- toward $-q$).
- Components **perpendicular** to **p** (along the bisector) are equal and opposite $\rightarrow$ cancel.
- Components **parallel** to **p** both point opposite to **p** and add: $E = 2E_+ \cos \theta$ with $\cos \theta = \frac{a}{\sqrt{r^2 + a^2}}$.
- $$E = \frac{2}{4\pi\epsilon_0} \frac{q}{r^2 + a^2} \cdot \frac{a}{(r^2 + a^2)^{1/2}} = \frac{1}{4\pi\epsilon_0} \frac{2qa}{(r^2 + a^2)^{3/2}} = \frac{1}{4\pi\epsilon_0} \frac{p}{(r^2 + a^2)^{3/2}}$$

FINAL RESULT

$$E_{\text{eq}} = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \quad (\text{opposite to } \mathbf{p}, r \gg a)$$

Limits: short dipole for the simplified form.

How to write it in the exam: State clearly that the perpendicular components cancel; many students lose marks by not saying why only the parallel components add.

Torque on a dipole in a uniform electric field

3 MARKS

Principle A pair of equal, opposite forces (a couple) produces rotation but no translation.

Assumptions Uniform field **E**; dipole makes angle θ with **E**.

- Force on $+q$: qE along **E**; on $-q$: qE opposite. Net force = 0.
- Perpendicular distance between the two forces: $2a \sin \theta$.
- Torque
 $\tau = (qE) (2a \sin \theta) = (2aq) E \sin \theta = pE \sin \theta$.

FINAL RESULT

$$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}, \quad \tau = pE \sin \theta$$

Limits: uniform field only. In a non-uniform field there is also a net force. τ is maximum (pE) at $\theta = 90^\circ$ and zero at 0° (stable equilibrium) and 180° (unstable equilibrium).

How to write it in the exam: Mention 'net force is zero, only torque acts' — it is a standard marking point.

1.6 Electric flux and Gauss's theorem

F FLUX AND GAUSS'S THEOREM

Electric flux through an area: $\Phi = \mathbf{E} \cdot \mathbf{A} = EA \cos \theta$ ($\text{N m}^2 \text{C}^{-1}$). The area vector $\mathbf{A}$ is normal to the surface (outward for a closed surface).

Gauss's theorem (statement): the net flux through any closed surface equals $1/\epsilon_0$ times the net charge enclosed.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

$\mathbf{E}$ is the total field of all charges. Flux depends only on enclosed charge, not on the surface shape. Always true; **handy for finding $\mathbf{E}$ only with symmetry.**

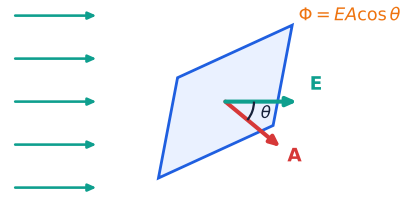


Fig. 1.7 Flux through a tilted area: only the component of $\mathbf{E}$ along $\mathbf{A}$ counts, $\Phi = EA \cos \theta$.

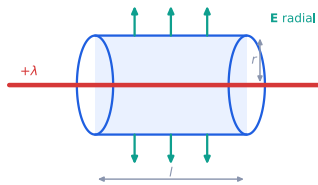


Fig. 1.8 Cylindrical Gaussian surface around a long wire; flux only through the curved part.

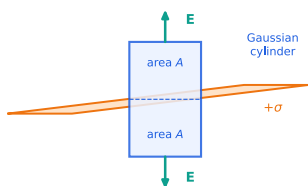


Fig. 1.9 Pillbox across an infinite sheet; flux through both end faces.

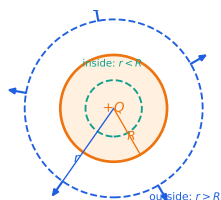


Fig. 1.10 Concentric spheres: $r > R$ encloses Q ; $r < R$ encloses nothing.

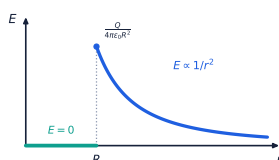


Fig. 1.11 E versus r for a thin charged shell.

A. Infinitely long straight charged wire

GAUSS

Principle Gauss's theorem with a coaxial cylinder.

Assumptions Uniform linear density λ ; wire very long, so by symmetry $\mathbf{E}$ is radial.

- Take a coaxial cylinder of radius r , length l . Flux through each flat end is zero ($\mathbf{E} \perp d\mathbf{A}$).
- Through the curved surface $\mathbf{E} \parallel d\mathbf{A}$ and E is constant: $\Phi = E (2\pi r l)$.
- Enclosed charge = λl . So $E (2\pi r l) = \lambda l / \epsilon_0$.

FINAL RESULT

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (\text{radially outward for } \lambda > 0)$$

Limits: far from the ends ($r \ll \text{length}$). $E \propto 1/r$.

How to write it in the exam: Draw the cylinder, state the symmetry argument, flux only through the curved part.

B. Infinite uniformly charged plane sheet

GAUSS

Principle Gauss's theorem with a small cylindrical 'pillbox'.

Assumptions Surface density σ ; by symmetry $\mathbf{E}$ is perpendicular to the sheet and the same distance on both sides.

- Pillbox of end-area A placed symmetrically; flux through the curved side is zero.
- Flux through the two ends: $EA + EA = 2EA$.
- Enclosed charge = σA , so $2EA = \sigma A / \epsilon_0$.

FINAL RESULT

$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{independent of distance from the sheet})$$

Limits: ideal infinite sheet. Two opposite sheets: $E = \sigma / \epsilon_0$ between, 0 outside.

How to write it in the exam: Two faces give the factor 2.

C. Uniformly charged thin spherical shell

GAUSS

Principle Gauss's theorem with concentric spherical surfaces.

Assumptions Total charge Q spread uniformly on a shell of radius R ; by symmetry $\mathbf{E}$ is radial and depends only on r .

- Outside ($r > R$):** $E (4\pi r^2) = Q / \epsilon_0 \Rightarrow E = \frac{Q}{4\pi\epsilon_0 r^2}$ — the shell acts like a point charge Q at its centre.
- On the surface ($r = R$):** $E = \frac{Q}{4\pi\epsilon_0 R^2} = \frac{\sigma}{\epsilon_0}$.
- Inside ($r < R$):** the Gaussian sphere encloses no charge, so $E (4\pi r^2) = 0 \Rightarrow E = 0$.

FINAL RESULT

$$E = 0 \quad (r < R), \quad E = \frac{Q}{4\pi\epsilon_0 r^2} \quad (r \geq R)$$

Limits: thin shell; E jumps by σ / ϵ_0 across the surface.

How to write it in the exam: Give both regions and sketch the E - r graph.

1.7 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$F = k \frac{ q_1 q_2 }{r^2}$	F (N), q (C), r (m); $k = 9 \times 10^9$	point charges at rest; direction: repulsive for like, attractive for unlike	$q_1 = q_2 = 1 \mu\text{C}$, $r = 0.1$ m: $F = 0.9$ N
$\mathbf{E} = \frac{kq}{r^2} \hat{\mathbf{r}}$	E (N C ⁻¹)	outward for $q > 0$, inward for $q < 0$	$q = 5$ nC, $r = 0.3$ m: $E = 500$ N/C
$\mathbf{F} = q\mathbf{E}$	F (N)	same direction as $\mathbf{E}$ for $q > 0$	e in 10^4 N/C: $F = 1.6 \times 10^{-15}$ N
$p = 2aq$	p (C m)	from $-q$ to $+q$	$2a = 1$ cm, $q = 10$ nC: $p = 10^{-10}$ C m
$E_{ax} = \frac{2kp}{r^3}$	$r \gg a$	along $\mathbf{p}$	$p = 10^{-10}$, $r = 0.1$: 1800 N/C
$E_{eq} = \frac{kp}{r^3}$	$r \gg a$	opposite to $\mathbf{p}$	half the axial value
$\tau = pE \sin \theta$	τ (N m)	$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$; uniform field	$\theta = 30^\circ$: $\tau = pE/2$
$\Phi = EA \cos \theta$	Φ (N m ² C ⁻¹)	θ between $\mathbf{E}$ and $\mathbf{A}$	$\mathbf{E} \parallel \mathbf{A}$: $\Phi = EA$
$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\epsilon_0}$	$\epsilon_0 = 8.854 \times 10^{-12}$	any closed surface	cube, charge q at centre: $q/6\epsilon_0$ per face
$E = \frac{\lambda}{2\pi\epsilon_0 r}; \frac{\sigma}{2\epsilon_0}$	wire ; sheet	infinitely long / infinite	sheet field is distance-independent

N1 C01-N01 Ch 1 Easy Numerical BOARD-STYLE PRACTICE 2 marks

Two point charges $4 \mu\text{C}$ and $6 \mu\text{C}$ are 30 cm apart in air. Find the force between them. What happens if they are placed in a medium of dielectric constant 4?

GIVEN $q_1 = 4 \times 10^{-6}$ C, $q_2 = 6 \times 10^{-6}$ C, $r = 30$ cm = 0.30 m

FIND Force F (air) and F' (medium)

FORMULA $F = \frac{kq_1 q_2}{r^2}$, $F' = F/\epsilon_r$

WORKING $F = \frac{9 \times 10^9 \times 4 \times 10^{-6} \times 6 \times 10^{-6}}{(0.30)^2} = \frac{0.216}{0.09} = 2.4$ N $\rightarrow F' = \frac{2.4}{4} = 0.6$ N

Answer: $F = 2.4$ N (repulsive); $F' = 0.6$ N

▲ Convert cm to m before squaring — a slip here changes the answer by 10^4 .

N2 C01-N02 Ch 1 Medium Numerical BOARD-STYLE PRACTICE 3 marks

Charges $+9$ nC and -9 nC are 20 cm apart. Find the electric field at the mid-point of the line joining them.

GIVEN $q = 9 \times 10^{-9}$ C; each charge is 10 cm = 0.10 m from the mid-point

FIND $\mathbf{E}$ at the mid-point

FORMULA $E = \frac{kq}{r^2}$ for each; fields add (same direction)

WORKING $E_+ = E_- = \frac{9 \times 10^9 \times 9 \times 10^{-9}}{(0.10)^2} = 8100$ N/C. Away from $+q$ and toward $-q$ are the same direction, so
 $E = 8100 + 8100 = 16200$ N/C.

Answer: 1.62×10^4 N C⁻¹ directed from $+9$ nC toward -9 nC

▲ Do not subtract: at the mid-point of opposite charges the two fields add.

N3 C01-N03 Ch 1 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A dipole has charges ± 5 nC separated by 2 cm. Find its dipole moment and the field at an axial point 10 cm from its centre.

GIVEN $q = 5 \times 10^{-9}$ C, $2a = 0.02$ m, $r = 0.10$ m

FIND p and E_{axial}

FORMULA $p = q(2a)$, $E = \frac{2kp}{r^3}$ (since $r \gg a$)

WORKING $p = 5 \times 10^{-9} \times 0.02 = 1.0 \times 10^{-10}$ C m. $E = \frac{2 \times 9 \times 10^9 \times 10^{-10}}{(0.1)^3} = \frac{1.8}{10^{-3}} = 1.8 \times 10^3$ N/C.

Answer: $p = 1.0 \times 10^{-10}$ C m; $E \approx 1.8 \times 10^3$ N C⁻¹ along $\mathbf{p}$

▲ Check: the exact formula gives 1.84×10^3 N/C — the $r \gg a$ approximation is good to about 2%.

1.8 Solved numericals (continued) and MCQ practice

N5 C01-N04 Ch 1 Hard Numerical BOARD-STYLE PRACTICE 3 marks

An infinite sheet carries uniform surface charge density $2 \mu\text{C m}^{-2}$. A thin spherical shell of radius 10 cm carries $8 \mu\text{C}$. Find (a) the field near the sheet, (b) the field of the shell at 20 cm and 5 cm from its centre.

GIVEN $\sigma = 2 \times 10^{-6} \text{ C m}^{-2}$; $Q = 8 \times 10^{-6} \text{ C}$, $R = 0.10 \text{ m}$

FIND E for the sheet; E (0.20) and E (0.05) for the shell

FORMULA sheet: $E = \frac{\sigma}{2\epsilon_0}$; shell: $E = \frac{kQ}{r^2}$ outside, 0 inside

WORKING (a) $E = \frac{2 \times 10^{-6}}{2 \times 8.854 \times 10^{-12}} = 1.13 \times 10^5 \text{ N/C}$. (b) at $r = 0.20 \text{ m}$: $E = \frac{9 \times 10^9 \times 8 \times 10^{-6}}{(0.20)^2} = 1.8 \times 10^6 \text{ N/C}$; at $r = 0.05 \text{ m}$ (inside): $E = 0$.

Answer: (a) $1.13 \times 10^5 \text{ N/C}$ (b) $1.8 \times 10^6 \text{ N/C}$ outside, 0 inside

▲ Inside a charged thin shell the field is zero for any point — no calculation needed.

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C01-M01 Ch 1 Easy Conceptual BOARD-STYLE PRACTICE

1. A glass rod rubbed with silk becomes positively charged because

- (a) electrons are transferred from silk to the rod
- (b) protons are transferred from the rod to silk
- (c) electrons are transferred from the rod to silk
- (d) protons are transferred from silk to the rod

Ans: (c) Only electrons (the loosely bound charge carriers) move; the rod loses electrons and becomes positive.

C01-M02 Ch 1 Easy Numerical BOARD-STYLE PRACTICE

2. Each of two point charges is doubled and the distance between them is also doubled. The force between them becomes

- (a) $F/4$
- (b) $F/2$
- (c) F
- (d) $2F$

Ans: (c) $F \propto q_1 q_2 / r^2 \rightarrow (2 \cdot 2) / (2^2) = 1$, so the force is unchanged.

C01-M03 Ch 1 Medium Conceptual

BOARD-STYLE PRACTICE

3. Two electric field lines cannot cross because

- (a) field lines are straight
- (b) the electric field at a point would have two directions
- (c) charges would be neutralised
- (d) the lines represent forces on neutral particles

Ans: (b) At the crossing point the tangent — hence $\mathbf{E}$ — would have two directions, which is impossible.

C01-M04 Ch 1 Medium Application

BOARD-STYLE PRACTICE

4. For a short dipole, the ratio of the field at an axial point to that at an equatorial point at the same distance r from the centre is

- (a) 1:2
- (b) 1:1
- (c) 2:1
- (d) 4:1

Ans: (c) $E_{ax} = 2kp/r^3$ and $E_{eq} = kp/r^3$.

C01-M05 Ch 1 Medium Numerical BOARD-STYLE PRACTICE

5. A dipole of moment $2 \times 10^{-8} \text{ C m}$ lies at 30° to a uniform field $5 \times 10^3 \text{ N/C}$. The torque is

- (a) $5 \times 10^{-5} \text{ N m}$
- (b) $1 \times 10^{-4} \text{ N m}$
- (c) $2.5 \times 10^{-5} \text{ N m}$
- (d) $8.7 \times 10^{-5} \text{ N m}$

Ans: (a) $\tau = pE \sin 30^\circ = 2 \times 10^{-8} \times 5 \times 10^3 \times 0.5 = 5 \times 10^{-5} \text{ N m}$.

C01-M06 Ch 1 Medium Application

BOARD-STYLE PRACTICE

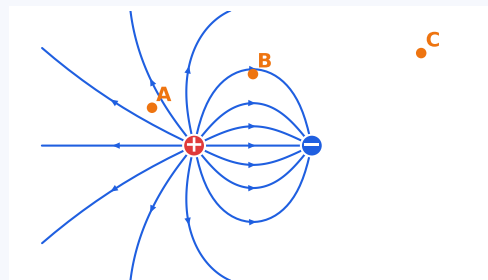
6. A point charge q is placed at the centre of a cube. The flux through one face is

- (a) q/ϵ_0
- (b) $q/4\epsilon_0$
- (c) $q/6\epsilon_0$
- (d) $6q/\epsilon_0$

Ans: (c) Total flux q/ϵ_0 is shared equally by six identical faces.

C01-M07 Ch 1 Medium Diagram BOARD-STYLE PRACTICE

7. In the dipole field-line figure, the field is strongest at point



- (a) A
- (b) B
- (c) C
- (d) equal at all three

Ans: (a) Lines are closest together near the +q charge, and line density is proportional to $|\mathbf{E}|$.

C01-AR01 Ch 1 Medium Assertion-Reason

BOARD-STYLE PRACTICE

8. Assertion (A): Gauss's law holds only for charge distributions with spherical symmetry.

Reason (R): Gauss's law is used to find $\mathbf{E}$ only when symmetry makes E constant over the Gaussian surface.

(a) A and R true, R explains A; (b) A and R true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a
- (b) b
- (c) c
- (d) d

Ans: (d) Gauss's law is valid for every closed surface and every distribution; symmetry is needed only to use it for calculating E .

R Chapter 1 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

- Field lines: $+q$, $-q$, dipole, two equal $+q$ (mark the neutral point)
- Dipole with P on the axial line and Q on the equatorial line, with $\mathbf{E}$ arrows
- Dipole in a uniform field showing the two forces and the arm $2a \sin \theta$
- Cylindrical Gaussian surface (wire) and pillbox (sheet)
- Concentric Gaussian spheres for a shell, and the E - r graph

★ 7-DAY REVISION PLAN

Day 1	Coulomb's law, superposition numericals
Day 2	Field, field lines, dipole concept
Day 3	Dipole derivations (axial, equatorial, torque)
Day 4	Gauss's theorem — three applications
Day 5	Numericals N1–N5 without notes
Day 6	MCQs + assertion–reason
Day 7	Redraw all diagrams; self-test

Chapter 1 at a glance — Electric Charges and Fields

Charge

Quantised $q = ne$; conserved; additive (scalar).

Coulomb's law

$F = kq_1q_2/r^2$; vector form with $\hat{r}$; superposition for many charges.

Field and lines

$\mathbf{E} = \mathbf{F}/q_0$; lines start +, end –; never cross; density $\propto E$.

Dipole

$p = 2aq$ (– to +); $E_{ax} = 2kp/r^3$ along $\mathbf{p}$; $E_{eq} = kp/r^3$ against $\mathbf{p}$.

Torque

$\tau = pE \sin \theta$; net force zero in uniform E ; stable at 0° .

Gauss

$\oint \mathbf{E} \cdot d\mathbf{A} = q_{enc}/\epsilon_0$; wire $\lambda/2\pi\epsilon_0 r$; sheet $\sigma/2\epsilon_0$; shell 0 inside, kQ/r^2 outside.

Formula map

$$F = \frac{kq_1q_2}{r^2}$$

Coulomb

$$E = \frac{kq}{r^2}$$

point charge

$$\mathbf{F} = q\mathbf{E}$$

force on charge

$$p = 2aq$$

dipole moment

$$\frac{2kp}{r^3}$$

axial

$$\frac{kp}{r^3}$$

equatorial

$$\tau = pE \sin \theta$$

torque

$$\frac{\lambda}{2\pi\epsilon_0 r}$$

wire

$$\frac{\sigma}{2\epsilon_0}$$

sheet

! COMMON MISTAKES

- Forgetting that E_{eq} is **opposite** to $\mathbf{p}$.
- Using $2\epsilon_0$ for a shell (it is ϵ_0 in Q/ϵ_0).
- Treating field lines as particle paths.
- Not converting nC, μC , cm to SI.
- Quoting $E = \sigma/2\epsilon_0$ for a conductor surface (it is σ/ϵ_0).

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can state and use Coulomb's law in vector form
- ☐ I can draw field lines for 4 configurations
- ☐ I can derive axial and equatorial dipole fields
- ☐ I can derive torque on a dipole
- ☐ I can apply Gauss's law to wire, sheet and shell
- ☐ I solved N1–N5 without looking
- ☐ I attempted all 10 MCQs

✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

- Charge on 5 electrons? | **Ans:** $-8.0 \times 10^{-19} \text{ C}$
- What is E inside a charged thin shell? | **Ans:** Zero
- $E_{ax} : E_{eq}$ for a short dipole? | **Ans:** 2:1
- SI unit of flux? | **Ans:** $\text{N m}^2 \text{ C}^{-1}$
- Flux through a closed surface enclosing no net charge? | **Ans:** Zero